

Calibrated Conversational Orchestration for Human-Supervised Agricultural Robotic Systems

GreenFieldData MSCA Joint Doctorate — PhD Position L

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Abstract. Precision agricultural robotics creates an operational paradox: the most capable field systems remain inaccessible to the agronomists and operators who must supervise them. This proposal investigates **AgriTalk**, a conversational AI system for human-supervised robotic orchestration in precision pesticide spraying. Conversational interaction is not a UI convenience: agricultural commands are contextually under-specified, temporally conditional, and collaborative by nature, requiring iterative disambiguation and trust negotiation. Three scientific questions structure the work: **(RQ1)** how calibrated uncertainty can translate NL commands into statistically grounded Human-in-the-Loop (HITL) trigger policies; **(RQ2)** how layer-wise Belief Vector Field (BVF) geometry can provide verifiably faithful command explanation — contrasting with Chain-of-Thought traces which reflect learned behaviour rather than computational belief; **(RQ3)** how streaming IoT field state can be atomically integrated into conversational grounding under real-time sensor uncertainty. Three primary contributions (C1–C3) target NLP, agricultural AI, and interpretability venues; a fourth (C4) provides Metaflow-based reproducible infrastructure as a scientific necessity for this multi-site, non-stationary PhD environment.

Introduction and Research Context

The GreenFieldData Programme and Position L. The GreenFieldData MSCA Joint Doctorate (UCBLyon1 + UniMI) addresses Sustainable Agriculture through data-driven precision robotics. Position L targets Natural Language Interfaces for Robotic and Analytical Farming Systems, supervised by Dr. Vargas-Solar (distributed data management, DAG-based computation) and Prof. Oberti (precision crop protection, agricultural robotics) [1]. I propose **AgriTalk**: a calibrated conversational orchestration system for precision pesticide spraying.

Why Conversational AI is Scientifically Appropriate Here. Agricultural commands are structurally under-specified: “*Spray more pesticide on the struggling section near the east boundary*” contains simultaneous ambiguity over spatial reference, chemical identity, dosage baseline, and phenological timing (whether a crop is at a stage susceptible to the chosen formulation). This is not a failure of language; it reflects three deep properties of agricultural work: **contextual embeddedness** (meaning depends on current field state, including NDVI¹ readings, soil moisture, and weed pressure), **temporal conditionality** (safety depends on wind speed, spray drift risk, and phenological stage), and **collaborative trust** (operators retain and must retain situational authority). A conversational system that properly handles this uncertainty becomes a mechanism for cooperative planning between human expertise and robotic precision.

Focal Scenario. Operator: “*Spray more pesticide on the struggling section near the east boundary.*” Simultaneous ambiguities: spatial reference (which section?), quantitative baseline (“more” than what?), chemical identity (which formulation, given current pesticide resistance profiles and phenological stage?), spray drift legality (current wind speed?). A wrong interpretation risks crop destruction, regulatory violation, and pesticide resistance buildup. Yet over-confirmation causes alert fatigue and abandonment. The scientific challenge is resolving ambiguity with calibrated statistical confidence and verifiable explanation.

State of the Art and Research Gaps. Current task-oriented dialogue (TOD) systems [2, 3] use fixed slot schemas unable to accommodate the situational dependence of agricultural commands. LLM softmax confidence is systematically overconfident on out-of-distribution inputs [4]. Chain-of-Thought (CoT) prompting produces reasoning traces that *appear* explanatory but are increasingly recognised as behavioral artifacts rather than

¹NDVI: Normalized Difference Vegetation Index, a satellite/drone-derived measure of crop health; values < 0.35 indicate stress zones requiring precision intervention.

windows into model computation: Turpin et al. [5] demonstrate that CoT-produced rationales are unfaithful — models change their reasoning narrative without changing internal processing; Kambhampati et al. [6] argue that anthropomorphizing intermediate tokens as “reasoning traces” is not a harmless metaphor but a category error that distorts how we evaluate model cognition; and Oxford AIGI [7] show directly that CoT is not explainability. No existing conversational AI system integrates real-time streaming IoT context into command grounding at agronomic granularity [8]. Retrieving externally grounded facts via RAG [9] reduces hallucination but does not address calibration under streaming temporal context drift.

Open Research Gaps:

1. No principled mapping from LLM uncertainty to statistically guaranteed HITL policy for safety-critical agricultural commands
2. No method for verifiably faithful (vs. plausible) explanation of NL-commanded agricultural decisions
3. No conversational AI system providing temporally consistent, atomic field-state grounding under heterogeneous IoT conditions
4. No empirical framework for measuring trust calibration improvement via faithful explanation in agricultural HITL user studies
5. No reproducible evaluation infrastructure accounting for non-stationarity (seasonal, farm, environmental drift) in agricultural AI

Research Questions and Contributions

Three primary research contributions + one infrastructure contribution:

- C1 [AI SAFETY] Calibrated Intent Compiler (CIC):** Conformal prediction-based NL-to-command compilation with statistically guaranteed HITL trigger policies. *Target: EMNLP/ACL 2027; ECPA workshop as initial venue.*
- C2 [AI INTERPRETABILITY] Belief-Grounded Explanation System (BGES):** Layer-wise Belief Vector Field (BVF) geometry for verifiably faithful command explanation — “learning to believe, not to behave” — validated against a human operator study. *Target: ACL/EMNLP 2028; FAccT 2029 for human study results.*
- C3 [NLP / CONV-AI] Temporal Streaming Grounding Aggregator (TSGA):** Atomically consistent, time-indexed field-state register enabling real-time contextual grounding under sensor uncertainty. *Target: EDBT workshop / Computers & Electronics in Agriculture 2028.*
- C4 [INFRASTRUCTURE] Reproducible Agricultural AI Framework:** Metaflow-based pipeline for reproducible evaluation under seasonal and cross-farm non-stationarity. *Not a research question; a scientific necessity enabling C1–C3.*

Research Questions (RQ1–RQ3):

- RQ1 (AI Safety)** Can conformal prediction sets provide statistically valid coverage guarantees for agricultural intent classification under distribution shift, and do those guarantees translate into HITL trigger policies that measurably reduce unsafe autonomous action and operator alert fatigue simultaneously?
- RQ2 (AI Interpretability)** Does layer-wise BVF belief tracing of fine-tuned agricultural LLMs provide verifiably faithful command attribution — detectable by human operator study — compared with Chain-of-Thought traces which are shown to be behaviorally generated rather than computationally causal?
- RQ3 (NLP/Conversational AI)** How can a streaming temporal context aggregator enable real-time, atomically consistent grounding of NL agricultural commands against heterogeneous IoT field state, and what are the failure boundaries under degraded or delayed telemetry?

System Architecture: AgriTalk Information Boundaries

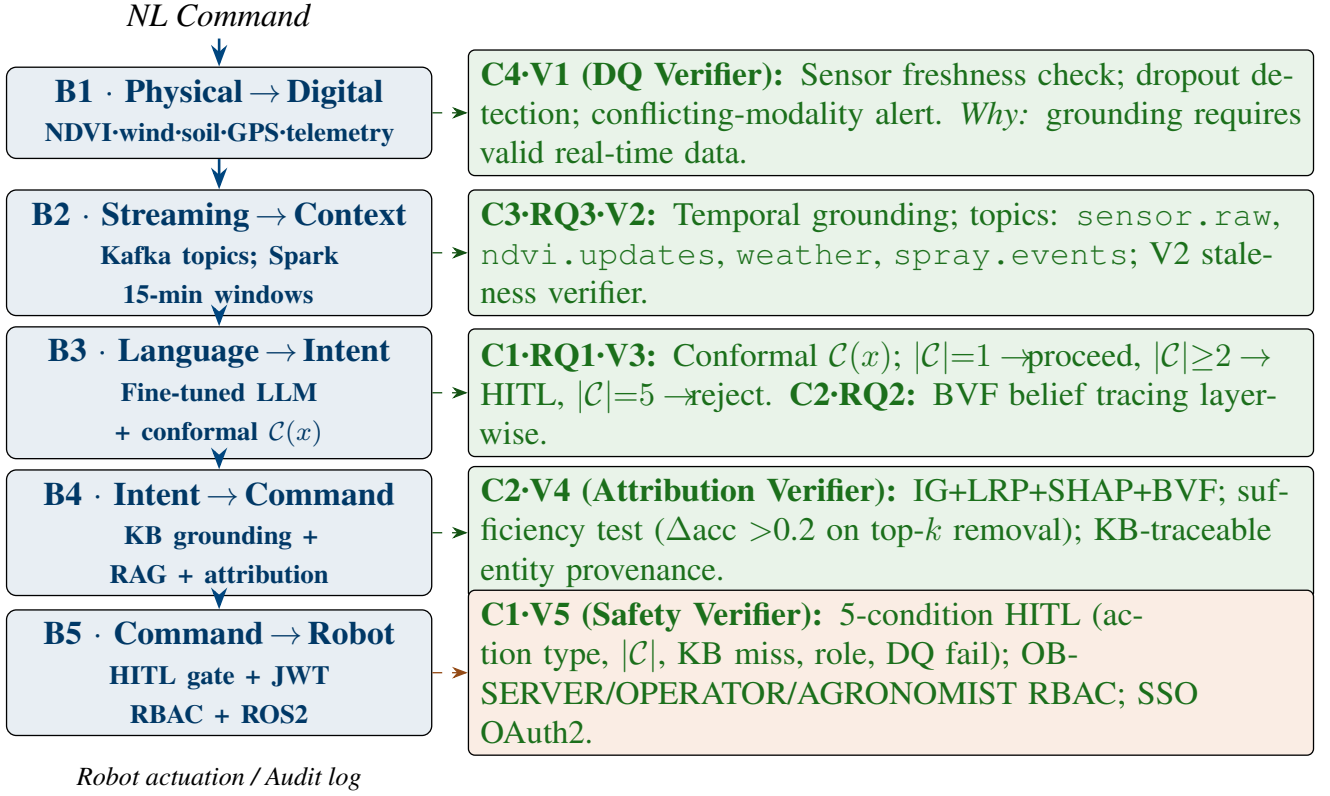


Figure 1: AgriTalk 5-Boundary Architecture. Left: information pipeline. Right: research contribution, verification layer, and design rationale per boundary. Dashed arrows indicate research loci.

Knowledge Base and Agronomic Grounding

Knowledge Base (KB) entries² encode agronomic constraints that cannot be inferred from NL alone: legal spray drift limits per formulation, safe application windows relative to phenological stage, pesticide resistance profiles, and NDVI-zone-indexed variable-rate application maps. RAG [9] retrieves these entries as grounding context; unverified entities are blocked at B4.

HITL Confirmation Gate and Role-Based Access Control

HITL³ gate triggers on any of five conditions: **(1)** Action type $\in \{\text{SPRAY}, \text{ABORT}, \text{DOSAGE_CHANGE}, \text{ZONE_OVERRIDE}, \text{HARVEST_EARLY}, \text{IRRIGATION_OVERRIDE}, \text{PESTICIDE_SWITCH}, \text{EMERGENCY_STOP}\}$; **(2)** $|\mathcal{C}(x)| \geq 2$ (conformal non-singleton); **(3)** any entity KB-unverified; **(4)** upstream DQ check failed (stale/missing sensor); **(5)** user role = OBSERVER. Intent classes: COMMAND, ABORT_MISSION, QUERY, REPORT, DIAGNOSE, CLARIFY, CONFIRM, DELEGATE.

Role	Auth Method	Command Access	HITL Requirement
FARMER/OBSERVER	SSO OAuth2/OIDC	QUERY, REPORT	All actions
FIELD_OPERATOR	SSO + MFA	COMMAND, DIAGNOSE	SPRAY, ABORT, critical
AGRONOMIST	SSO + MFA	Full command set	ABORT_MISSION only
AI_RESEARCHER	SSO + API key	Model/calib. inspection	N/A
ADMIN	SSO + hardware token	Full system	N/A

²KB entries: structured agronomic records from EPPO (European and Mediterranean Plant Protection Organisation) and PPDB (Pesticide Properties DataBase), encoding pesticide formulations, legal spray drift limits, crop-specific dosage protocols, resistance profiles, and AGROVOC-linked ontological constraints.

³HITL: Human-in-the-Loop — the deliberate inclusion of human judgment at critical decision points before AI-commanded actuator execution.

Metaflow as Scientific Infrastructure for Non-Stationary Agricultural AI

The Non-Stationarity Problem. Agricultural AI experiments are intrinsically non-stationary: weather, crop lifecycle, weed pressure, soil variation, and operator vocabulary differ across seasons, farms, and years. Reproducibility requires preserving not only models and code but *temporal environmental state*: sensor lineage, field context snapshots, calibration sets, and deployment conditions. Without this, comparing Paper 1 (C1, Lyon, Year 1) against Paper 2 (C3, Milan, Year 2) is scientifically unreliable — not a tool problem but a scientific validity problem.

The Three Vs, Mapped Technically:

- **Velocity:** Steps are independently re-runnable; iterating on the conformal calibration step (B3) does not re-trigger Kafka IoT ingestion (B1–B2). Concretely: `python agritalk_flow.py run -start-step=calibrate` replays from any saved artifact ID.
- **Validation:** Supervisors in Lyon and Milan inspect *identical* artifact-level run state via `metaflow.Run('AgriTalkFlow')` — no environment mismatch.
- **Versioning:** Every run is immutably tagged (run ID, timestamp, Git SHA, Metaflow artifact hash). Model regression is detected by comparing `AgriTalkFinetuneFlow` run IDs; calibration drift is tracked per seasonal checkpoint.

Seasonal Replay — a PhD-level feature: re-running any historical pipeline step with archived sensor snapshots to ask “*would the HITL policy calibrated in Lyon winter have triggered correctly on UniMI spring data?*” — answered with evidence, not assumption. This is why Metaflow is preferred over Airflow (no artifact versioning), Kubeflow (adds Kubernetes complexity without prototype-to-production continuity), or MLflow (tracks experiments but does not orchestrate pipelines).

Research Methodology and Evaluation

C1 (RQ1): Conformal Prediction and the Trajectory to Safe Autonomy

Standard softmax confidence [4] is overconfident on OOD agricultural commands. I replace it with conformal prediction sets [10, 11]:

$$\mathbb{P}(y \in \mathcal{C}(x)) \geq 1 - \alpha, \quad \alpha = 0.05$$

The trajectory from *unsafe* to *safe autonomy* is empirically measurable: an uncalibrated system executes commands with false confidence (unsafe); a conformal system escalates to HITL when genuinely uncertain. As calibration evidence accumulates across seasons, α is adaptively relaxed — expanding autonomous action while maintaining coverage. Mondrian conformal provides per-class guarantees, critical for the ABORT class.

Critical open question (RQ1 body): Does the exchangeability assumption hold under seasonal drift? We investigate adaptive conformal recalibration [11] with seasonal checkpoints; coverage degradation triggers conservative fallback (mandatory HITL for all critical actions). *V3 Verifier*: monitors rolling ECE and $|\mathcal{C}|$ distribution per 1000-query window; alerts when empirical coverage drops below nominal $1 - \alpha$ by > 0.02 .

C2 (RQ2): Belief Vector Field Geometry vs. Chain-of-Thought

Why Chain-of-Thought Is Not the Answer. CoT prompting generates a reasoning narrative that *appears* explanatory. Three bodies of evidence refute its sufficiency as explainability: **(a)** Turpin et al. [5]: CoT rationales are systematically unfaithful — models produce different reasoning chains for the same internal processing depending on prompt framing, and frequently predict the same answer regardless of whether the CoT “reasoning” is preserved or corrupted; **(b)** Kambhampati et al. [6]: intermediate tokens labelled as “reasoning traces” or “thoughts” are behavioral outputs, not computational records — anthropomorphizing them as transparent cognition is a category error that produces dangerously misleading system evaluations; **(c)** Oxford AIGI [7]: CoT is not explainability — it is a learned output distribution artifact. CoT also amplifies biases from fine-tuning data: a model trained on biased agronomic instructions produces authoritative CoT narratives that surface no uncertainty and cite no verifiable source. **This PhD investigates the alternative:** layer-wise

Belief Vector Field (BVF) geometry [12] traces how the model’s internal representational state evolves through depth — “*learning to believe, not to behave.*” At early layers: syntactic beliefs (what is the verb, what is the object?). At middle layers: semantic entity beliefs (“struggling section” → NDVI zone < 0.35?). At late layers: agronomic safety beliefs (is the dosage within EPPO legal limits?). This layer-by-layer depthwise signal is the actual computational pathway — not a post-hoc narrative — and is therefore verifiable in a controlled user study.

Attribution methods and verifiers. Five attribution methods are applied to the fine-tuned agricultural LLM, each with an associated *V4* verifier:

Method	Mechanism	Verifier (V4)
Integrated Gradients [13]	Gradient $\times$ input along interpolation path	Sufficiency: top- k removal $\Delta\text{acc} > 0.2$
LRP [14]	Layer-wise relevance backpropagation	Necessity: keep top- k , $\text{acc} > 0.80$
SHAP [15]	Shapley values over token coalitions	Cross-method: Kendall $\tau(\text{IG}, \text{LRP}) > 0.5$
Attention Rollout [16]	Normalized attention flow through heads	Consistency: rank correlation across layers
BVF Geometry (this work)	Layer curvature in representation space	Geometric stability across calibration runs

Human operator study (Year 2, UniMI): three explanation conditions: **(A)** no explanation; **(B)** CoT-generated plausible narrative (unverified); **(C)** faithful: rule-auditable trigger string + KB source URIs + BVF attribution. Measures: operator *trust calibration gap*, *intervention frequency*, *correction accuracy*, *decision latency*. Hypothesis: Condition C improves trust calibration without increasing cognitive load compared to B — tested, not assumed.

C3 (RQ3): Streaming Temporal Context Aggregator

The TSGA maintains a **field-state register** via Kafka topics: `agritalk.sensor.raw` (15-sec), `agritalk.ndvi.update` (per drone mission), `agritalk.weather.forecast` (10-min), `agritalk.robot.telemetry` (1-sec during ops), `agritalk.spray.events` (immutable audit; treatment traceability), `agritalk.alerts`. Spark Structured Streaming applies: 15-min rolling mean (sensors), 48-hr tumbling window (weather), 7-day lookback (spray history and pesticide resistance tracking). Grounding uses the *exact snapshot at query receipt time* — enabling conditional resolution: “Is it safe to spray *now*?” (current register) vs. “Was it safe Tuesday?” (Metaflow-archived artifact from that timestamp).

Failure boundary study (RQ3 body): explicit stress-testing under sensor dropout, delayed telemetry (>5-min lag), and conflicting modality inputs (GPS vs. vision positioning). *V2 Verifier*: monitors register freshness per topic; flags when any sensor channel exceeds a staleness threshold; triggers mandatory HITL and logs a DQ failure reason in the audit record.

Evaluation Framework (with Justification)

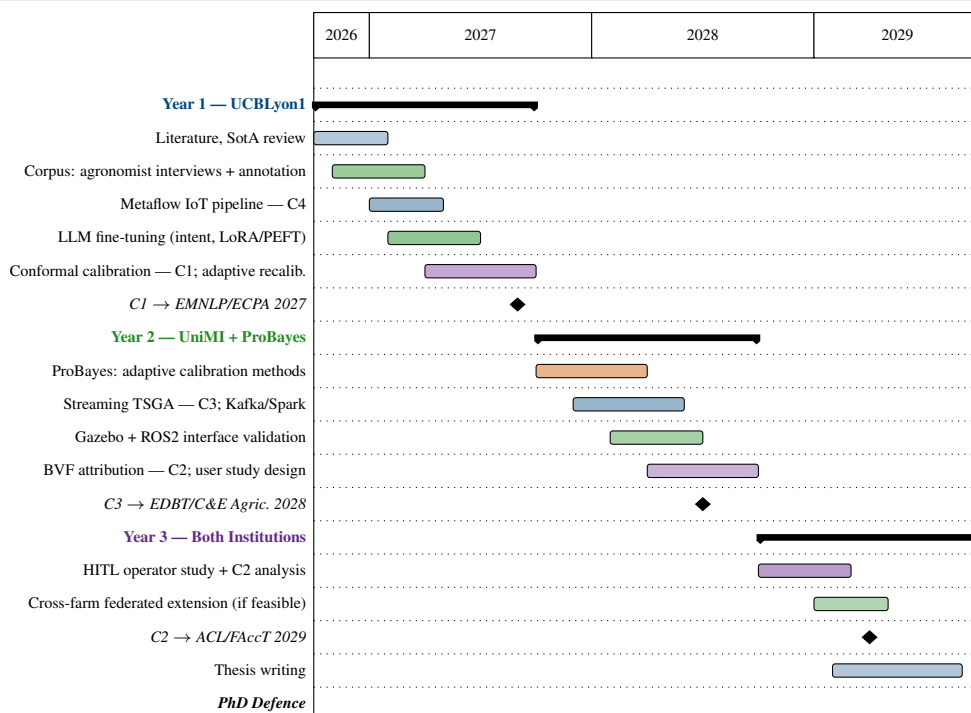
Metric	Justification for inclusion	Operational target
Macro-F1 (intent)	Multi-class imbalance; ABORT is rare but critical	Benchmark-informed [2]
ECE (calibration)	Measures calibration gap vs. softmax baseline [4]	< softmax baseline ECE
ABORT recall	Type-II error is safety-critical; asymmetric cost	Highest-priority metric
Coverage stability	Tracks calibration drift under seasonal distribution shift	Recalibration frequency
KB hit rate	Monitors KB coverage gap; alerts on new entity types	Distribution over farms
IG-LRP Kendall τ	Cross-method agreement $\Rightarrow$ stable attribution signal	$\tau > 0.5$
Sufficiency (Δacc)	Verifier V4: top- k removal should change prediction	$\Delta > 0.2$
Trust calibration gap	User study: C3 overconfidence vs. C1 no-explanation	Directional improvement
HITL fatigue rate	Alert fatigue baseline [17]	Below 20% on QUERY
Edge P95 latency	ABORT actuator safety requirement (edge deployment)	<800 ms on Jetson

Note: hard thresholds above are initial objectives informed by prior work; achievable values are determined empirically. Degradation below nominal triggers verifier alerts, not project failure.

Datasets and Corpus (Year-by-Year)

- **Year 1:** AgroNLP corpus (constructed via agronomist interviews, Lyon); PANGAEA soil/crop data [18]; HuggingFace gchhablani/bert-large-cased-finetuned-agri-ner (agricultural NER pre-training); MBZUAI/agriculture-llm (LLM fine-tuning base); ACRE precision agriculture benchmark; OpenWeather historical API (Lyon 2020–2026).
- **Year 2:** ProBayes partner farm data (proprietary, GDPR-compliant, federated); UniMI experimental farm robotic spraying logs (novel corpus); FieldRobot Event 2026/2027 simulation data; USDA-ARS-AgAID/farmland_agricult (HuggingFace); OpenWeather (Milan counterpart); EPPO/PPDB live API (regulatory KB).
- **Year 3:** Cross-farm federated corpus (2–4 farms, CFL [19]); EU Farm-to-Fork monitoring data (where available); synthetic adversarial corpus for edge-case robustness testing.

Three-Year Roadmap and Risk Mitigation



Publication note: venues above are targets; workshop and agricultural conference papers are first-class outputs. Three well-evaluated publications are the goal.

Risk	Primary Plan	Backup Plan
Real robot access delayed	Gazebo simulation (Months 1–18)	All 3 papers feasible on simulation + ACRE
Corpus too small	Agronomist interviews + expert-validated synthetic	MultiWOZ → agricultural transfer
Conformal coverage drifts	Adaptive recalibration; seasonal checkpoints	Conservative fixed threshold; full HITL
BVF attribution unstable	Cross-method verifier V4 flags early	SHAP + LRP as primary; BVF as analysis
Edge latency >800 ms	Phi-3-mini / Llama-3.2-3B distilled	Command-type cache for frequent patterns

Reflection, Design Choices, and Candidature

Why conversational AI over rule-based? Rules cannot scale to the vocabulary and compositional diversity of natural agricultural instruction; however safety-critical execution retains symbolic HITL safeguards. The research value is characterising *where* learned conversational AI adds over rule-based baselines. **Why conformal prediction over Bayesian?** Bayesian calibration requires prior specification and MCMC — not tractable for

800 ms edge inference. Conformal provides distribution-free coverage from a single calibration pass [11]. **Why BVF over CoT?** CoT is behavioral; BVF is causal. This is not a design preference but an empirically testable claim, addressed directly by RQ2. **Why open models?** Gradient access for attribution (C2); reproducibility; runs on UCBLyon1 GPU infrastructure. **Why simulation-first?** I am not a robotics hardware engineer. Year 1 Gazebo simulation develops and validates the ROS2 command interface. Real robot validation in Year 2 (under Prof. Oberti) builds on a proven interface. Honest scoping.

Why I am the right candidate. (a) **Belief Vector Field research** (ongoing, under review): experience with layer-wise LLM representation tracing and attribution limitations — including over-smoothing and saturation artefacts — provides the critical foundation for C2. (b) **CFL federated learning work** [19]: direct methodological basis for Year 3 cross-farm generalisation. (c) **Teaching and IoT (50+ student projects)**: practical understanding of sensor deployment and of explaining AI to non-specialist practitioners — directly applicable to the HITL user study design. (d) **Supervision fit**: Dr. Vargas-Solar → DAG-based data management and streaming query formalisation (C3, C4); Prof. Oberti → precision agriculture domain and field validation infrastructure (C1, C2 user study). The joint structure maps cleanly to my contributions.

The open questions I find most intellectually compelling are precisely the ones I cannot yet answer: Does conformal calibration remain valid under seasonal distribution shift? Can faithful BVF-grounded explanation measurably improve operator trust calibration, or does it overwhelm practitioners with irrelevant geometric detail? These questions are why this is a PhD and not an engineering project.

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